# CO-1 AT-2

## **Assignment Title:** Agile Sprint Planning & User Story Workshop


**Name:** Sk. Parvez Ahamed

**Reg. No:** 192411084

**Course Code:** CSA0117- Software Engineering

**Faculty Name**: Dr. Subramanian P


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## Blockchain-Based Academic Credential Verification Platform

## 1. Identify the Project Vision and Stakeholders

### 1.1 Project Vision

The Blockchain-Based Academic Credential Verification Platform is designed to provide a secure, transparent, and decentralized system for issuing, storing, and verifying academic certificates. The platform leverages blockchain technology to ensure that certificates cannot be altered, duplicated, or forged after issuance. By replacing traditional manual verification methods with an automated digital verification process, the platform increases trust among educational institutions, employers, students, and government authorities. The solution also improves efficiency, reduces administrative costs, and enables quick verification of credentials across multiple organizations.

### 1.2 Project Objectives

The proposed system has the following objectives:

- To issue secure digital academic certificates using blockchain technology.
- To eliminate fake certificates and document forgery.
- To enable instant verification of academic credentials through QR codes or certificate IDs.
- To provide a transparent and tamper-proof audit trail for every issued certificate.
- To support multiple universities and educational institutions on a common platform.
- To generate detailed verification reports for authorized users.
- To improve data integrity, security, and accessibility of academic records.

### 1.3 Key Stakeholders

| Stakeholder | Responsibilities |
| --- | --- |
| Universities | Register students, issue certificates, maintain academic records |
| Students | View, download, and securely share their digital certificates |
| Employers | Verify the authenticity of certificates before recruitment |
| Government Authorities | Monitor institutions and ensure compliance with educational standards |
| System Administrator | Manage users, institutions, permissions, and platform maintenance |
| Blockchain Network | Store certificate hashes securely and maintain immutable records |

### 1.4 Stakeholder Benefits

Each stakeholder benefits differently from the proposed system:

- Universities reduce paperwork, prevent certificate forgery, and simplify certificate issuance.
- Students gain lifelong access to secure digital credentials that can be shared instantly.
- Employers save time by verifying certificates within seconds instead of contacting universities.
- Government authorities can audit educational records efficiently and maintain institutional transparency.
- Administrators can monitor system performance, user activities, and security through centralized dashboards.

### 1.5 Project Scope

The Blockchain-Based Academic Credential Verification Platform is designed to serve educational institutions, students, employers, and government authorities. The system covers the complete lifecycle of academic credentials, including certificate creation, digital signing, blockchain storage, verification, report generation, and audit management. It supports multiple institutions while ensuring secure access through role-based authentication. The scope is limited to academic credential management and does not replace existing student information systems; instead, it integrates with them to improve verification efficiency.

## 2. Create Epics and User Stories

### Epic 1 – User Authentication and Access Management

This epic focuses on providing secure access to the platform for all authorized users.

**User Story 1**

As a student, I want to securely log into the platform so that I can access my academic credentials anytime.

**User Story 2**

As a university administrator, I want secure authentication so that only authorized personnel can issue certificates.

### Epic 2 – Certificate Issuance

This epic handles the creation and management of digital academic certificates.

**User Story 3**

As a university administrator, I want to create and issue digital certificates so that students receive authentic academic credentials.

**User Story 4**

As a student, I want to download my digital certificate so that I can share it with employers or universities.

### Epic 3 – Certificate Verification

This epic enables organizations to verify certificates quickly.

**User Story 5**

As an employer, I want to verify certificates using a QR code or Certificate ID so that I can recruit genuine candidates.

### Epic 4 – Reporting and Audit

**User Story 6**

As a system administrator, I want to generate verification reports and maintain audit logs so that all verification activities are traceable.

### 2.1 Product Roadmap

The project roadmap outlines the major milestones planned for the development of the platform.

| **Phase** | **Activities** |
|---|---|
| Phase 1 | Requirement gathering and stakeholder analysis |
| Phase 2 | UI/UX design using Figma |
| Phase 3 | Sprint planning and backlog creation |
| Phase 4 | User authentication and certificate management |
| Phase 5 | Blockchain integration |
| Phase 6 | Certificate verification and report generation |
| Phase 7 | Testing and deployment |

## 3. Define Acceptance Criteria

User Story 1

Given the student has a valid account

When the correct username and password are entered,

Then the student should successfully access the dashboard.

**User Story 3**

Given the university administrator has entered valid student details,

When the "Issue Certificate" button is clicked,

Then the system should generate a digital certificate, digitally sign it, store its hash on the blockchain, and generate a QR code.

**User Story 5**

Given a valid QR code or Certificate ID,

When the employer submits the verification request,

Then the system should retrieve the blockchain record, compare the certificate hash, and display the verification result.

**User Story 6**

Given certificate verification activities exist, When the administrator selects "Generate Report",

Then the system should produce a downloadable report containing verification history, timestamps, and user details.

## 4. Prioritize the Product Backlog

The Product Backlog contains all the features required to develop the system. High-priority items are selected first because they provide the greatest business value and are essential for the platform's core functionality.

| Product Backlog ID | Feature | Priority | Reason |
|---|---|---|---|
| PB-01 | User Registration & Login | High | Required for secure access |
| PB-02 | University Registration | High | Required before certificate issuance |
| PB-03 | Student Management | High | Core functionality |
| PB-04 | Digital Certificate Issuance | High | Primary feature |
| PB-05 | Blockchain Integration | High | Ensures security and immutability |
| PB-06 | QR Code Generation | High | Enables quick verification |
| PB-07 | Certificate Verification | High | Main business requirement |
| PB-08 | Verification Reports | Medium | Improves reporting |
| PB-09 | Audit Trail | Medium | Supports compliance |
| PB-10 | Dashboard Analytics | Low | Enhancement feature |

**5. Plan the Sprint Backlog**

**Sprint Duration**

**2 Weeks (14 Days)**

***Sprint Goal***

The goal of Sprint 1 is to build the core modules of the platform that enable secure user authentication, certificate issuance, blockchain integration, and certificate verification. Completing these features will establish the foundation for future enhancements.

**Sprint Backlog**

| User Story | Tasks |
|---|---|
| Authentication | Design login page, create database, implement authentication |
| Student Module | Create student registration, update profile, dashboard |
| University Module | Institution registration, approval workflow |
| Certificate Issuing | Create certificate template, digital signature integration |
| Blockchain Integration | Generate certificate hash, store on blockchain |
| QR Code Module | Generate unique QR code |
| Verification Module | Compare blockchain records and display verification result |
| Testing | Unit testing, integration testing, bug fixing |

**6. Estimate Effort Using Story Points**

The development team uses the Fibonacci sequence (1, 2, 3, 5, 8, 13, 21) to estimate the complexity and effort required for each user story. Larger story point values indicate greater development effort and uncertainty.

| User Story | Story Points | Justification |
|---|---|---|
| User Authentication | 3 | Basic login functionality with moderate complexity |
| Student Dashboard | 5 | Multiple interface components and database connectivity |
| University Registration | 5 | Includes institution validation and approval |
| Certificate Issuance | 8 | Requires digital signatures and certificate generation |
| Blockchain Integration | 13 | Most technically complex module |
| QR Code Generation | 5 | Requires certificate linking and generation |
| Certificate Verification | 8 | Blockchain comparison and validation logic |
| Report Generation | 3 | Standard report creation functionality |

**Why Story Points?**

Story points measure the overall effort required for implementing a feature rather than estimating exact development hours. They consider complexity, uncertainty, and implementation effort, making sprint planning more accurate and flexible.

**7. Present the Sprint Goal and Expected Deliverables**

***Sprint Goal***

The primary goal of Sprint 1 is to deliver a functional prototype of the Blockchain-Based Academic Credential Verification Platform with secure authentication, certificate issuance, blockchain integration, and certificate verification capabilities.

**Expected Deliverables**

At the end of Sprint 1, the following deliverables are expected:

- Secure User Registration and Login Module
- University Management Module
- Student Management Module
- Digital Certificate Issuance System
- Blockchain Hash Storage
- QR Code Generation
- Certificate Verification Module
- Verification Report Module
- Basic Administrator Dashboard
- Unit Test Cases and Testing Report
- Sprint Review Documentation

## Conclusion

The Agile Sprint Planning and User Story Workshop provides a structured roadmap for developing the Blockchain-Based Academic Credential Verification Platform. By identifying stakeholders, organizing requirements into epics and user stories, defining acceptance criteria, prioritizing the product backlog, estimating development effort using story points, and planning sprint activities, the project team can deliver high-quality software incrementally. This Agile approach encourages collaboration, continuous feedback, adaptability to changing requirements, and timely delivery of valuable features, ultimately resulting in a secure, scalable, and reliable academic credential verification system.